

CS-GY 6763: Lecture 2

Hashing + Fingerprinting, Chebyshev's Inequality

NYU, Prof. Ainesh Bakshi

Note on Mathematical Proofs

It can be hard to know how formal to be. We will try to provide feedback on first problem set for anyone who is either too rigorous or too loose. It's a learning process.

Things that are generally fine:

- Can assume input size n is $> C$ for some constant c . E.g.
 $n > 2, n > 10$.

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- Derivatives, integrals, etc. can be taken from e.g. WolframAlpha without working through steps.
- Basic inequalities can be used without proof, as long as you verify numerically. Don't need to include plot on problem set.

Example Inequality

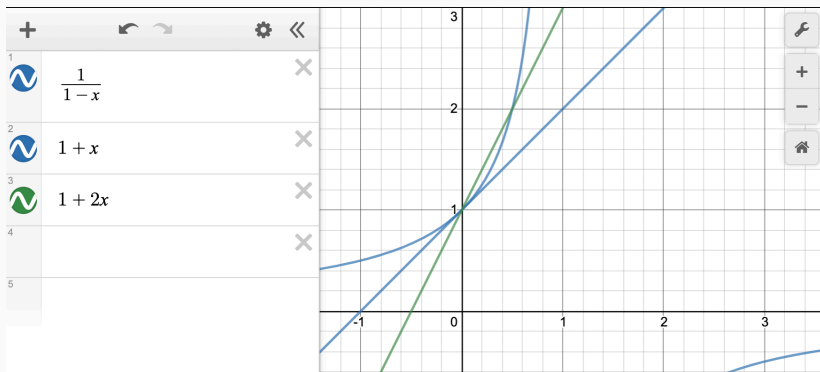
$$1 + \epsilon \leq \frac{1}{1 - \epsilon} \leq 1 + 2\epsilon \text{ for } \epsilon \in [0, .5].$$

Proof by plotting:

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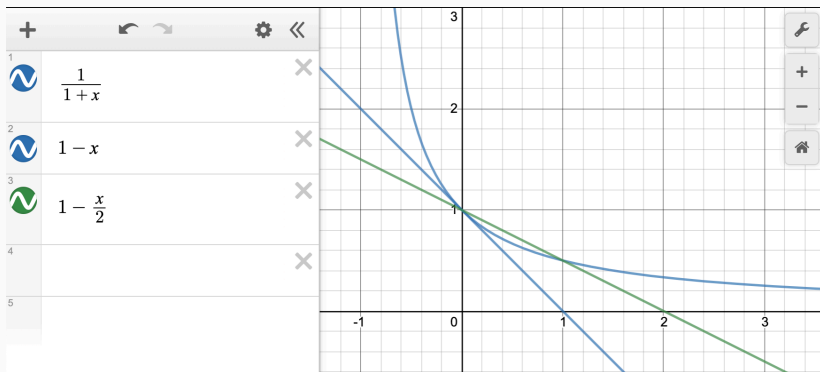
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- To start: $\frac{1}{n} - \frac{1}{n^2} \approx \frac{1}{n}$. Later: $\frac{1}{2n} \leq \frac{1}{n} - \frac{1}{n^2} \leq \frac{1}{n}$.

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- $\log(n/2) \approx \log(n)$ Later: $\log(n)/2 \leq \log(n/2) \leq \log(n)$.

Definitions of Independence

Suppose we have random variables $X_1, \dots, X_k$. We say that **a pair of random variables** X_i and X_j are independent if, for all possible values v_i, v_j ,

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Does mutual independence imply pairwise independence? What about the converse?

Definitions of Independence

Give an example of three random variables that are pairwise independent but not mutually independent.

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Example: Let $X \sim \text{Uni}\{-1, 1\}$, $Y \sim \text{Uni}\{-1, 1\}$ and $Z = XY$.

Linearity of Variance

If we have two independent random variables X, Y , then:

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y].$$

If we have a set of pairwise independent¹ random variables $X_1, \dots, X_k$ then:

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Proof Sketch:
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Mutual independence is not necessary!

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Uniformly Random Hash Function

Let $h : \mathcal{U} \rightarrow \{1, \dots, m\}$ be a random function. This means that h is constructed by an algorithm using a seed of random numbers, but then the function is fixed.

Recall: Uniformly Random Hash Function. A random function $h : \mathcal{U} \rightarrow \{1, \dots, m\}$ is called uniformly random if:

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 - Which implies that $\Pr[h(x) = h(y)] = \frac{1}{m}$

$$\Pr[h(x) = h(y) = h(z)] = \sum_{i \in [m]} \Pr[h(x) = i, h(y) = i, h(z) = i]$$

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$$\begin{aligned}\Pr[h(x) = h(y) = h(z)] &= \sum_{i \in [m]} \Pr[h(x) = i, h(y) = i, h(z) = i] \\ &= \sum_{i \in [m]} \frac{1}{m^3} = \frac{1}{m^2}.\end{aligned}$$

Uniformly Random Hash Function

The only way to implement a truly random hash function is to create a giant lookup table, where the numbers on the right are chosen independently at random from $\{1, \dots, m\}$.

x	$h(x)$
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If we're hashing 35 char ASCII strings (e.g. urls) the length of the table is greater than the number of atoms in the universe.

Universal Hash Functions

For the application to CountMin from last class we can weaken our assumption that h is uniformly random.

Definition (Universal hash function)

A random hash function $h : \mathcal{U} \rightarrow \{1, \dots, m\}$ is universal if, for any fixed $x, y \in \mathcal{U}$,

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Claim: A uniformly random hash-function is universal.

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How much space does this hash function take to store?

Limited Independence Hash Functions

Similar alternative definition:

Definition (Pairwise independent hash function)

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Proof: $\Pr[h(x) = h(y)] = \sum_{i \in [m]} \Pr[h(x) = i \text{ and } h(y) = i] = \sum_{i \in [m]} \frac{1}{m^2} = \frac{1}{m}.$

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Pseudorandom Hash Functions

We won't prove that random polynomials provide good hash functions, but I want to give a flavor of what is involved (e.g., why do prime numbers show up?).

Fingerprinting

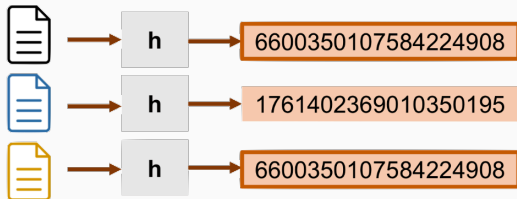
Goal: Construct a compact “fingerprint” $h(f)$ for any file f with two properties:

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- The fingerprints $h(f_1)$ and $h(f_2)$ should be different with high probability if the contents of f_1 and f_2 differ at all.
- If the contents of f_1 and f_2 are identical, we should have $h(f_1) = h(f_2)$.



(Basically the same goal as most applications of hashing.)

Applications of Fingerprinting

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- Check that a file pieced together from multiple parts is not missing anything.

Goal: Construct a compact “fingerprint” function $h(f)$ such that:

- $h(f_1) \neq h(f_2)$ if $f_1 \neq f_2$ with high probability.

Ideally, length of $h(f_1)$ (i.e. the size of the integers hashed to) is much less than the file size.

Random Fingerprinting

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$$h(f) = f \pmod{p}.$$

How many bits does $h(f)$ take to store?

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$$\log(p) = O(\log(tn \log(tn))) = O(\log(n) + \log(t)).$$

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Claim: If $f_1 \neq f_2$ then $h(f_1) = h(f_2)$ with probability $\leq \frac{2}{t}$.

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Claim: If $f_1 \neq f_2$ then $h(f_1) = h(f_2)$ with probability $\leq \frac{2}{t}$.

Since our fingerprint only takes $O(\log n + \log t)$ space, we can set t to be super large, so effectively the probability of $h(f_1)$ and $h(f_2)$ colliding is negligible for all real-world applications.

E.g. set fingerprint length to $\log n + 28$ bits and you are more likely to win Megamillions.

Random Fingerprinting

How do we sample a random prime between $2, \dots, tn \log n$?

Keep in mind that n is pretty large here. For a 200kb image, $n \approx 1.6$ million.

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So each iteration is efficient, but is this efficient overall?

Roughly how many tries do you expect this to take?

Prime Number Theorem

Let $\pi(x)$ denote the number of primes less than some integer x .

Informally:

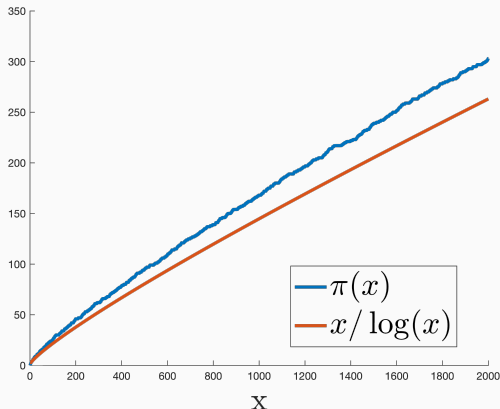
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Formally: For $x > 17$,

$$\frac{x}{\ln(x)} \leq \pi(x) \leq \frac{x}{\ln(x) - 4}$$

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After a few hundred tries, we will almost definitely find a prime number. **In general, need $O(q)$ tries in expectation to find a prime with q bits.**

Remark: Finding large prime numbers is important in some other applications beyond hashing.

Random Fingerprinting

$$h(f) = f \pmod{p} \quad \text{for prime } p \in \{2, \dots, tn \log(tn)\}$$

Claim: If $f_1 \neq f_2$ then $h(f_1) = h(f_2)$ with probability $\leq \frac{2}{t}$.

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First observation: If $h(f_1) = h(f_2)$, then:

In other words, we only fail if $|f_1 - f_2|$ is divisible by p .

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First observation: If $h(f_1) = h(f_2)$, then:

$$(f_1 - f_2) \pmod{p} = 0.$$

In other words, we only fail if $|f_1 - f_2|$ is divisible by p .

Question: What is the chance that $|f_1 - f_2|$ is divisible by a random prime $p \in \{2, \dots, tn \log(tn)\}$?

Number of distinct prime factors of $|f_1 - f_2|$: At most n .

Random Fingerprinting

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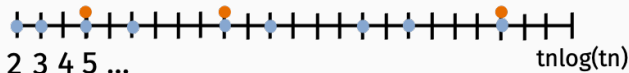
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- = prime number
- = prime factors of $f_1 - f_2$



Chance we pick a prime factor of $f_1 - f_2$ is less than:

$$\frac{n}{\frac{tn \log(tn)}{\log(tn \log(tn))}} = \frac{\log(tn \log(tn))}{t \log(tn)} \leq \frac{2 \log(tn)}{t \log(tn)}$$

Random Fingerprinting

Conclusion: The chance that a random prime $p \in \{2, \dots, tn \log(tn)\}$ is a factor of $|f_1 - f_2|$ is $\leq \frac{2}{t}$.

So, for two files $f_1 \neq f_2$, the chance that $h(f_1) = h(f_2) \leq \frac{2}{t}$.

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Set $t = 2^{28}$ (the chance you win Megamillions).

Fingerprint size: At most $2 \log_2(nt) = 2 \log_2(n) + 2 \log_2(2^{28})$ bits.

Suppose we are fingerprinting 200kb image files. $n \approx 1,600,000$, so our fingerprint has size:

96 bits

This amounts to a 17,000x reduction over sending and comparing the original files.

Remainder of Lecture

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This week we will discuss two more tools:

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Next week:

- Union Bound + Exponential Tail Bounds



**These six tools combined are surprisingly powerful and flexible.
They form the cornerstone of randomized algorithm design.**

Chebyshev's Inequality

A new concentration inequality:

Lemma (Chebyshev's Inequality)

Let X be a random variable with expectation $\mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}[X]$. Then for any $k > 0$,

$$\Pr[|X - \mathbb{E}[X]| \geq k \cdot \sigma] \leq \frac{1}{k^2}$$

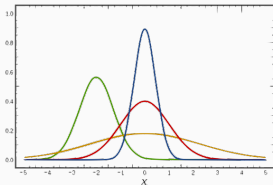
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$\sigma = \sqrt{\text{Var}[X]}$ is the standard deviation of X . Intuitively this bound makes sense: it is tighter when σ is smaller.

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Properties of Chebyshev's inequality:

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- **Bad/Good:** Requires a bound on the variance of X .

No hard rule for which to apply! Both Markov's and Chebyshev's are useful in different settings.

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Idea: Apply Markov's inequality to the (non-negative) random variable $S = (X - \mathbb{E}[X])^2$.

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Quick Example

If I flip a fair coin 100 times, show that with $> 93\%$ chance I get between 30 and 70 heads.

Let $C_1, \dots, C_{100}$ be independent random variables that are 1 with probability $1/2$, 0 otherwise.

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$$\Pr[|H - 50| \geq 20] \leq \frac{25}{20^2} = \frac{1}{16} = 6.25\%.$$

Applications of Chebyshev's Inequality

Abstract architecture of a streaming algorithm:

Have massive dataset $X = x_1, \dots, x_n$ with n pieces of data that arrive in a sequential stream. There is far too much data to store or process it in a single location.

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- To do so, we must compress data “on-the-fly”, storing some smaller data structure which still contains interesting information.
- Often can only take a single-pass over the data.

Count-Min was our first example of a streaming algorithm for the (ϵ, k) -frequent items problem.

Streaming Algorithms in Practice

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Lots of software frameworks exist for easy development of streaming algorithms.

Distinct Elements Problem

Input: $x_1, \dots, x_n \in \mathcal{U}$ where $\mathcal{U}$ is a huge universe of items.

Output: Number of distinct inputs.

Example: $f(1, 10, 2, 4, 9, 2, 10, 4) \rightarrow 5$

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- Number of distinct queries to a search engine.
- Distinct motifs in DNA sequence.

Implementations widely used at Google (Sawzall, Dremel, PowerDrill), Twitter, Facebook (Presto), etc.

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Goal: Return $\tilde{D}$ satisfying

$$(1 - \epsilon)D \leq \tilde{D} \leq (1 + \epsilon)D$$

with high probability and using only $O(1/\epsilon^2)$ space.

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- Choose random hash function $h : \mathcal{U} \rightarrow [0, 1]$.

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The hash function h maps from $\mathcal{U}$ to a random point in $[0, 1]$?

Hashing to real numbers:

- Impossible to implement $h(x)$ in reality, but you can replace it with $\frac{g(x)}{k}$, where g is a hash function that maps to $\{0, 1, \dots, k\}$ for sufficiently large k .

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- Just like when we assumed uniform random hash functions, this is a useful abstraction which makes understanding and analyzing algorithms easier.

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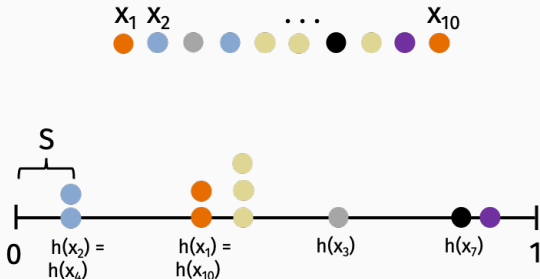
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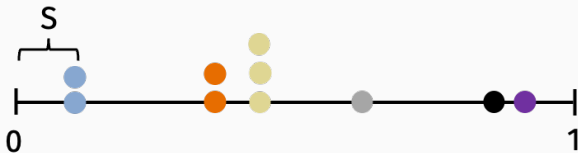
Visualization

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Visualization

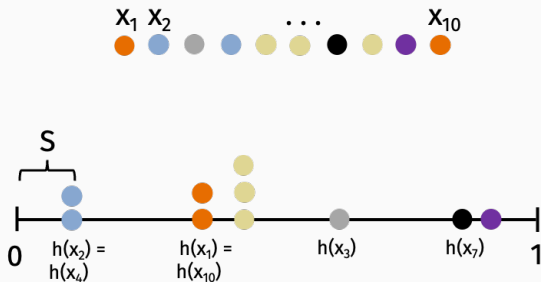


Important: If $\mathcal{U} \rightarrow [0, 1]$ uniformly at random, we can assume that there are no collisions. If we instead used a discrete grid, it would suffice to use a hash table of size $O(D^2)$ or, conservatively, $O(|\mathcal{U}|^2)$.

We will not do a formal analysis, but roughly how many bits does S take to store?

FM Analysis

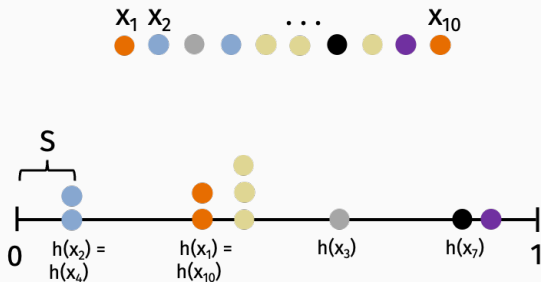
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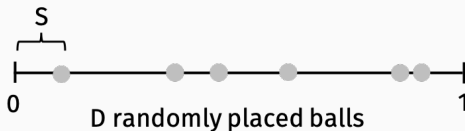
Intuition: When D is larger, S will be smaller. Makes sense to return the estimate $\tilde{D} = \frac{1}{S} - 1$.

What is $\mathbb{E}S$?

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Let D equal the number of distinct elements in our stream.

Lemma

$$\mathbb{E}S = \frac{1}{D+1}.$$

The Calculus Proof

Proof:

$$\mathbb{E}[S] = \int_0^1 \Pr[S \geq \lambda] d\lambda$$

Exercise: Why?

The Calculus Proof

Proof:

$$\begin{aligned}\mathbb{E}[S] &= \int_0^1 \Pr[S \geq \lambda] d\lambda \\ &= \int_0^1 (1 - \lambda)^D d\lambda\end{aligned}$$

Exercise: Why?

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Exercise: Why?

Hint: For a non-negative random variable $X = \int_0^\infty \mathbf{1}(X \geq t) dt$.
Then,

$$\mathbb{E}[X] = \mathbb{E}\left[\int_0^\infty \mathbf{1}(X \geq t) dt\right] = \int_0^\infty \Pr[X \geq t] dt.$$

Proving Concentration

$\mathbb{E}S = \frac{1}{D+1}$. **Estimate:** $\tilde{D} = \frac{1}{S} - 1$. We have for $\epsilon < \frac{1}{4}$:

If $(1 - \epsilon)\mathbb{E}S \leq S \leq (1 + \epsilon)\mathbb{E}S$, then:

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So, it suffices to show that S concentrates around its mean. I.e. that

$$|S - \mathbb{E}S| \leq \epsilon \cdot \mathbb{E}S.$$

Recall:

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$$1 - \epsilon \leq \frac{1}{1 + \epsilon} \leq 1 - .5\epsilon \text{ for } \epsilon \in [0, 1].$$

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Inverting the inequalities,

$$\frac{1}{(1 + \epsilon)\mathbb{E}S} \leq \frac{1}{S} \leq \frac{1}{(1 - \epsilon)\mathbb{E}S}$$

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Using $\mathbb{E}S = \frac{1}{D+1}$,

$$\frac{D+1}{1 + \epsilon} \leq \frac{1}{S} \leq \frac{D+1}{1 - \epsilon}$$

$$\Rightarrow (1 - \epsilon)D + (1 - \epsilon) - 1 \leq \frac{1}{S} - 1 \leq (1 + 2\epsilon)D + (1 + 2\epsilon) - 1$$

$$\Rightarrow (1 - \epsilon)D - \epsilon \leq \tilde{D} \leq (1 + 2\epsilon)D + 2\epsilon$$

□

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What should we compute about S next in order to apply Chebyshev?

Lemma

$$\text{Var}[S] = \mathbb{E}[S^2] - \mathbb{E}[S]^2 = \frac{2}{(D+1)(D+2)} - \frac{1}{(D+1)^2} \leq \frac{1}{(D+1)^2}.$$

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www.wolframalpha.com/input?i=antiderivative+of+%281-sqrt%28x%29%29%5ED

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Recall we want to show that, with high probability,
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Chebyshev's: $\Pr[|S - \mu| \geq \epsilon\mu] \leq \Pr[|S - \mu| \geq \epsilon\sigma] \leq \frac{1}{\epsilon^2}$.

Vacuous bound. Our variance is way too high!

Trick of the trade: Repeat many independent trials and take the mean to get a better estimator.

Given i.i.d. (independent, identically distributed) random variables $X_1, \dots, X_n$ with mean μ and variance σ^2 , what is:

- $\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \mu$

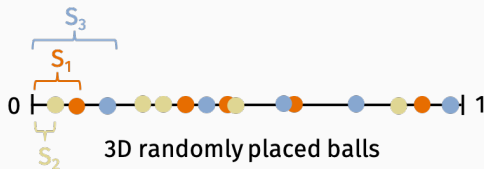
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FM Analysis

Using independent hash functions, maintain k independent sketches $S_1, \dots, S_k$.

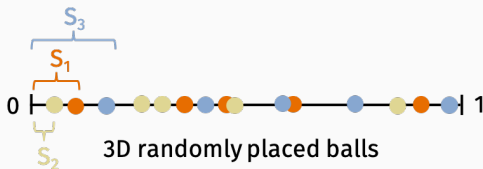


Flajolet–Martin:

- Choose k random hash functions $h_1, \dots, h_k : \mathcal{U} \rightarrow [0, 1]$.

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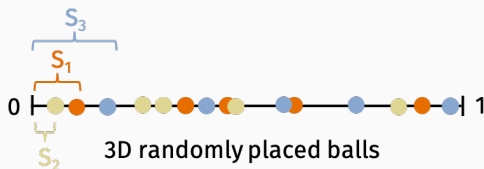


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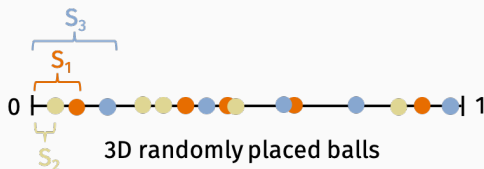


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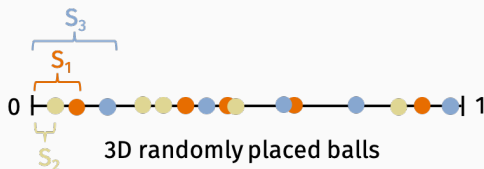


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- Return: $\frac{1}{S} - 1$

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Total space complexity: $O\left(\frac{1}{\epsilon^2 \delta}\right)$ to estimate distinct elements up to error ϵ with success probability $1 - \delta$.

- Recall that to ensure $(1 - \bar{\epsilon})D \leq \frac{1}{5} - 1 \leq (1 + \bar{\epsilon})D$, we needed $|S - \mu| \leq \frac{\bar{\epsilon}}{4}\mu$.
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- Recall that to ensure $(1 - \bar{\epsilon})D \leq \frac{1}{5} - 1 \leq (1 + \bar{\epsilon})D$, we needed $|S - \mu| \leq \frac{\bar{\epsilon}}{4}\mu$.
- So apply the result from the previous slide with $\epsilon = \bar{\epsilon}/4$.
- Need to store $k = \frac{1}{\epsilon^2\delta} = \frac{1}{(\bar{\epsilon}/4)^2\delta} = \frac{16}{\bar{\epsilon}^2\delta}$ counters.

Note on Failure Probability

$O\left(\frac{1}{\epsilon^2\delta}\right)$ space is an impressive bound:

- $1/\epsilon^2$ dependence cannot be improved.

²Technically, if we account for the bit complexity of storing $S_1, \dots, S_k$ and the hash functions $h_1, \dots, h_k$, the space complexity is $O\left(\frac{\log D}{\epsilon^2\delta}\right)$.

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We can get a better bound depending on $O(\log(1/\delta))$ using exponential tail bounds. We will see next lecture.

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